

Jaeyoung Kim

Ph.D. Student

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Summary

Ph.D. student at Seoul National University researching Homogeneous Dynamics, Ergodic Theory, and dynamics on Lie groups & buildings under the supervision of Prof. Seonhee Lim.

Education

Graduate Research Assistant	Seoul National University (SNU) , Homogeneous Dynamics (PI: Seonhee Lim)	Seoul, South Korea Aug 2022 – present
Integrated M.S. / Ph.D. Course	Seoul National University (SNU) , Mathematical Sciences • Advisor: Prof. Seonhee Lim (DASOM Lab) • Research Focus: Homogeneous Dynamics & Ergodic Theory	Seoul, South Korea Mar 2022 – present
Research Internship	Seoul National University (SNU) , Quantum Information Theory (PI: Sang-Gyun Youn)	Seoul, South Korea June 2020 – Mar 2021
Bachelor of Science (B.S.)	Seoul National University (SNU) , Mathematical Education • Leave for Mandatory Military Service (1/2018 – 9/2019)	Seoul, South Korea Mar 2016 – Feb 2022

Publications & Preprints

Irreducibly $SU(2)$-covariant quantum channels of low rank Euijung Chang, Jaeyoung Kim, Hyesun Kwak, Hun Hee Lee, Sang-Gyun Youn (Reviews in Mathematical Physics, 34(07), 2250021)	Jan 2022
Markov chain of diagonal action on the standard quotient of the building associated to $PGL_3(F_q((t^{-1})))$ Jaeyoung Kim, Sanghoon Kwon, Seonhee Lim (Work in progress)	2026+
Distribution of periodic torus orbits on Siegel modular variety and Duke’s theorem for matrix-valued quadratic form Jaeyoung Kim (Work in progress)	2026+

Talks

Markov chain of diagonal action on the standard quotient of the building associated to $PGL_3(F_q((t^{-1})))$ Poster Session, Beyond Uniform Hyperbolicity, Abdus Salam International Centre for Theoretical Physics (ICTP)	Trieste, Italy June 2025
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Markov chain of diagonal action on the standard quotient of the building associated to $\mathrm{PGL}_3(\mathbb{F}_q((t^{\frac{1}{p}}-1)))$ Special Session, 2024 KMS Annual Meeting, Korean Mathematical Society (KMS)	Suwon, South Korea Oct 2024
Dynamical property for horospherical action on the standard quotient of the building $\mathrm{B}_3(\mathbb{F}_q((t^{\frac{1}{p}}-1)))$ Lightning Talk, Dynamical Group Theory III: New Methods in Group Actions on Manifolds, KIAS	Seoul, South Korea Aug 2024
Dynamical property for horospherical action on the standard quotient of the building $\mathrm{B}_3(\mathbb{F}_q((t^{\frac{1}{p}}-1)))$ Lightning Talk, Introductory school: Rigidity, Dynamics and Geometric Structures, CIRM	Marseille, France Apr 2024
Non-escape of mass for horospherical action on the standard quotient of the building $\mathrm{B}_3(\mathbb{F}_q((t^{\frac{1}{p}}-1)))$ under type 1 geodesic flow Poster Session, 2023 KMS Annual Fall Meeting, KMS	SNU, Seoul, South Korea Oct 2023

Presentations & Seminar Talks

Work of Lindenstrauss, Margulis, Mohammadi, Shah: Effective Avoidance Principle and Diophantine Condition Research Seminar on Homogeneous Dynamics	Apr 2025
Work of Cantat, DuJardin: Hyperbolic dynamics on Compact Kähler Surfaces Course: Topics in Topology	Dec 2024
Quasi-Fuchsian groups and Bers' Simultaneous Uniformization theorem SNU Graduate Course: Differential Geometry 2	Nov 2024
Sullivan's No Wandering Domain Theorem SNU Graduate Course: Differential Geometry 2	Oct 2024
Markov chain of diagonal action on the standard quotient of the building associated to $\mathrm{PGL}_3(\mathbb{F}_q((t^{\frac{1}{p}}-1)))$ Exchange Seminar	Oct 2024
Iteration of Polynomial-like maps SNU Graduate Course: Differential Geometry 2	Sept 2024
The Scope of Dirichlet forms: Closability and Examples Course: Topics in Mathematical Methods of Probability	Apr 2024
Bruhat-Tits tree of SL_2 over a local field Research Seminar on Homogeneous Dynamics, SNU	Mar 2024
Work of Ciobotaru, Finkelshtein, Sert: Non-escape of mass and Equidistribution for Horospherical actions on Trees Exchange Seminar, SNU	Aug 2023
Work of Michel, Venkatesh: Equidistribution, L-functions and Ergodic theory Course: Topics in Topology	June 2023

Seminars

Exchange Seminar Supervisor: Gyeseon Lee	SNU 2023-01 - Present
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Reading Seminar on Arithmetic of Hyperbolic 3-manifolds

SNU
2023-09 - 2024-08

Research Seminar on Homogeneous Dynamics

Supervisor: Seonhee Lim

SNU
2022-09 - Present

Conferences Attended

2026 Algebra Camp

Chuncheon, South Korea
Jan 2026

Dynamical Group Theory VII

SNU, Seoul, South Korea
Nov 2025

2025 KMS Annual Meeting

KMS, Seoul, South Korea
Oct 2025

Dynamical Group Theory VI

KIAS, Seoul, South Korea
Sept 2025

Numeration and Substitution 2025

University of Tsukuba,
Japan
Sept 2025

2025 2K-GATE One day Geometric Topology Festival

KIAS, Seoul, South Korea
Aug 2025

A Summer Dynamics Day

Dongguk University, Seoul,
South Korea
Aug 2025

One-day Festival of Group actions, Geometry and Dynamics

KIAS, Seoul, South Korea
Aug 2025

Geometry and Dynamics in Higher Rank Lie Groups

SLMath, Berkeley, USA
July 2025

Beyond Uniform Hyperbolicity

ICTP, Trieste, Italy
June 2025

2025 KMS Spring Meeting

KMS, Daejeon, South Korea
Apr 2025

Workshop on Homogeneous and Complex Dynamics 2025

SNU, Seoul, South Korea
Mar 2025

KIAS-LFANT Winter School on Number Theory

KIAS, Seoul, South Korea
Jan 2025

2024 KMS Annual Meeting

KMS, Suwon, South Korea
Oct 2024

Dynamical Group Theory IV: KIAS-Rice Workshop on Geometric Topology

HCMC of KIAS, Seoul, South
Korea
Sept 2024

Dynamical Group Theory III: New Methods in Group Actions on Manifolds	KIAS, Seoul, South Korea Aug 2024
Number Theory Intensive Lecture Series: Introduction to p-adic Hodge theory	POSTECH, Pohang, South Korea June 2024
Introductory school: Rigidity, Dynamics and Geometric Structures	CIRM, Marseille, France Apr 2024
2024 Algebra Camp	Yangpyeong, South Korea Feb 2024
Winter School on Homogeneous Dynamics	SCMS, Fudan University, Shanghai, China Jan 2024
Hyperbolic Geometry of Numbers	HCMC of KIAS, Seoul Dec 2023
2023 KMS Annual Fall Meeting: KMS Special Conference with 2022 Fields Medalists	KMS, Seoul Oct 2023
Summer School on Arithmetic Statistics	KIAS, Seoul Aug 2023
2023 KMS Spring Meeting	KMS, Daejeon Apr 2023
Workshop on Dynamics of Group Action	Gangneung, South Korea Apr 2023
QSMS 2023 Winter Workshop on Number Theory and Representation Theory	QSMS, Jeonju, South Korea 2023-01 - 2023-02

Teaching

TA: Differential and Integral Calculus Practice 1 (003) & 2 (005)	SNU Spring & Fall 2025
TA: Calculus Practice 2 (001, 026)	SNU Spring & Fall 2024
TA: Honor Calculus Practice 2 (006)	SNU Spring 2023
Grader: Number Theory (001)	SNU Spring 2023
TA: Calculus Practice 1 (021)	SNU Spring 2023
TA: Calculus Practice 1 (022) & 2 (009)	SNU Spring & Fall 2022

Teaching Practice

Seoul National University
Middle School
May 2021

Teaching Volunteer

Guhyun High School
Jan 2020

Honors & Fellowships

Graduate Student Instructor Scholarship

Seoul National University
2022-03 - Present

BK21 Research Fellowship

Seoul National University
2022-03 - 2023-02, 2023-09 -
2025-02, 2025-09 - Present

Next-Generation Scholars in Fundamental Academic Disciplines

Seoul National University
2023-03 - 2023-08

Languages

English

B2 level

Korean

Native

Extra

Military Service

Army sergeant, Discharge after Completing Mandatory Service (Korean Army)

2018-01 - 2019-09